

Group Projects

Three open projects. Your group picks one, finds its own data, and answers a question that matters in your own field of engineering. There is no model answer and no dataset handed to you: choosing well is part of the work.

HOW IT WORKS

- **Groups of five or six**, from the same programme where possible.
- **Pick one project** from the three below. Groups in the same class may pick the same project, but not the same data.
- **Collect your own data.** Measure it, or take it from a published source you can cite. A minimum of **30 observations**; more is better for Projects 2 and 3.
- **Hand in a report** of 8 to 12 pages, plus your data file and any code.
- **Present for 10 minutes**, with 5 minutes of questions. Every member speaks. I will ask at least one person to explain a calculation.

PROJECT 1. MEASURE SOMETHING, AND SAY HOW SURE YOU ARE

Describing variation and estimating a parameter

Every measurement carries error. Quantify it.

THE QUESTION

Choose one quantity in your field that varies from unit to unit or from day to day. Measure it properly, describe how it varies, and give an interval estimate for its true mean or proportion with a stated confidence level. Then say what that interval means for a decision an engineer would take.

WHAT TO DO

1. Define the population precisely. "Concrete on site" is not a population; "35 MPa cubes cast on Block B between weeks 1 and 4" is.
2. Choose and justify a sampling procedure from Chapter 1. Say why you did not use the others.
3. Collect at least 30 observations. Record when, where and how.
4. Summarise with measures of location, dispersion and skewness. Draw the distribution.
5. Build a confidence interval for the mean, or for a proportion if your quantity is pass/fail.
6. State what would have to change for the interval to be half as wide.

WHERE TO FIND YOUR QUANTITY

Mechanical	Surface roughness across a batch; fuel consumption per trip; bolt torque at installation
Computer science	API response time; build duration; battery drain per hour of the same task
Electrical	Mains voltage through the day; LED luminous output across a batch; standby current draw
Civil	Concrete cube strength at 28 days; traffic counts in a fixed hour; slump test values
Geological	Assay grade along a drill core; rock density by sample; fracture spacing
Geomatic	GPS positional error at a known point; levelling misclosure; repeated total-station distances
Environmental	PM _{2.5} at a roadside; water turbidity; daytime noise level

The trap. Most groups collect 30 numbers from one machine, one site, or one hour, then claim an interval for the whole population. If your sample cannot support the claim, say so. Honest limits earn more marks than a confident wrong answer.

PROJECT 2. MODEL THE UNCERTAINTY, THEN DECIDE

Fitting a distribution and using expectation to choose

A distribution is only useful if it changes what you do.

THE QUESTION

Choose a quantity whose behaviour is uncertain, fit a probability distribution to it, and use that model to make a decision with money, risk or safety attached. Your recommendation must come from an expected value calculation, not from opinion.

WHAT TO DO

1. Argue from the mechanism which distribution should apply before you look at the fit. Counts of rare events, time to failure and a sum of many small effects each point somewhere different.
2. Fit it. Compare the fitted curve against your data honestly.
3. Compute probabilities that matter: the chance of exceeding a threshold, of failing before warranty, of a queue passing capacity.
4. Set up at least two courses of action with costs, and compute the expected cost or benefit of each using Chapter 5.
5. Recommend one, and state the assumption that would most change your answer if it were wrong.

WHERE TO FIND YOUR QUANTITY

Mechanical	Bearing life before failure; spare-part demand per month; time between breakdowns
Computer science	Requests arriving per minute; time between system faults; cache miss rate
Electrical	Outages per month on a feeder; transformer load peaks; component lifetime
Civil	Vehicle arrivals at a junction; daily rainfall against a drain's capacity; concrete strength against a specified minimum
Geological	Grade above a cut-off; blast fragment size; number of intersections above a threshold per hole
Geomatic	Residual errors in an adjustment; survey rework rate; time to fix a GNSS ambiguity
Environmental	Days exceeding a WHO air quality limit; spill events per year; effluent pH outside consent

The trap. Fitting a normal distribution to everything. Failure times and event counts are usually not normal, and forcing them there will give you a positive probability of a negative lifetime. Notice that and explain it.

PROJECT 3. TEST A CLAIM, AND FIND A RELATIONSHIP

Hypothesis testing and regression on real data

Does the thing people say on site actually hold?

THE QUESTION

Find a claim that people in your field repeat as though it were settled. Test it. Then model how your outcome depends on something you can measure, and say how much of the variation you have actually explained.

WHAT TO DO

1. State the claim, then write it as a null and an alternative hypothesis before collecting data.
2. Choose the right test from Chapter 7 and justify it. Say what you assumed and whether your data support the assumption.
3. Report the test statistic, the decision and what it means in plain language. Do not write "significant" without saying significant of what.
4. Fit a regression relating your outcome to a predictor. Report the coefficients, their interpretation in real units, and the correlation.
5. Plot the residuals and say what they show about the model.
6. State one thing your model must not be used for.

CLAIMS WORTH TESTING

Mechanical	"The new lubricant lowers running temperature"; does load predict wear rate?
Computer science	"The refactor made it faster"; does input size predict runtime?
Electrical	"Voltage is worse in the evening"; does load predict harmonic distortion?
Civil	"This mix is stronger than the old one"; does curing time predict strength?
Geological	"Grade is higher on the east side"; does depth predict density?
Geomatic	"Longer occupations give better fixes"; does baseline length predict error?
Environmental	"Air is worse near the highway"; does traffic count predict $PM_{2.5}$?

The trap. Treating a strong correlation as proof that one thing causes the other. Rainfall and ice-cream sales move together too. If you cannot rule out a third factor, say which one worries you.

THE PRESENTATION

Ten minutes, five for questions. Aim it at engineers in another programme: if only your own class can follow it, it is too narrow.

- Open with the decision or question, not with your method.
- One slide on how the data were collected. Be honest about what went wrong.
- Show the data before you show the model.
- Every member speaks, and should be able to answer on any part.
- Finish with what you would do differently with twice the data.

HOW IT IS MARKED

COMPONENT	MARKS	WHAT EARNS THEM
Problem and data	20	A question worth asking, a population defined precisely, sampling justified, collection documented honestly
Statistical work	30	Right method for the question, assumptions stated and checked, calculations correct
Interpretation	25	What the numbers mean in engineering terms, limits acknowledged, a decision actually reached
Report	10	Clear structure, readable figures with labelled axes, sources cited
Presentation	15	Clarity, timing, and how the group handles questions

Marks are awarded to the group, except where questioning shows a member cannot explain the work they are credited with.

DATA HONESTY

Submit the data you actually collected. A small, messy, honestly described dataset scores better than a clean invented one, and invented data is usually obvious: it is too well behaved. If a measurement went wrong, keep

it in and explain it. If you use a published dataset, cite it and say who collected it and why.

WHAT TO HAND IN

- Report as a single PDF, 8 to 12 pages including figures.
- Your data as a spreadsheet or CSV.
- Any code you wrote.
- Slides as a PDF.
- A one-line statement of who did what, signed by every member.

Upload through the submission link on the course page. Do not email your work.